pfi

Python for Fixed Income

COURSE 1 · PYTHON FOUNDATIONS

Files and Folders

WEEK 1 · DAY 2

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The desk lead's fourth note

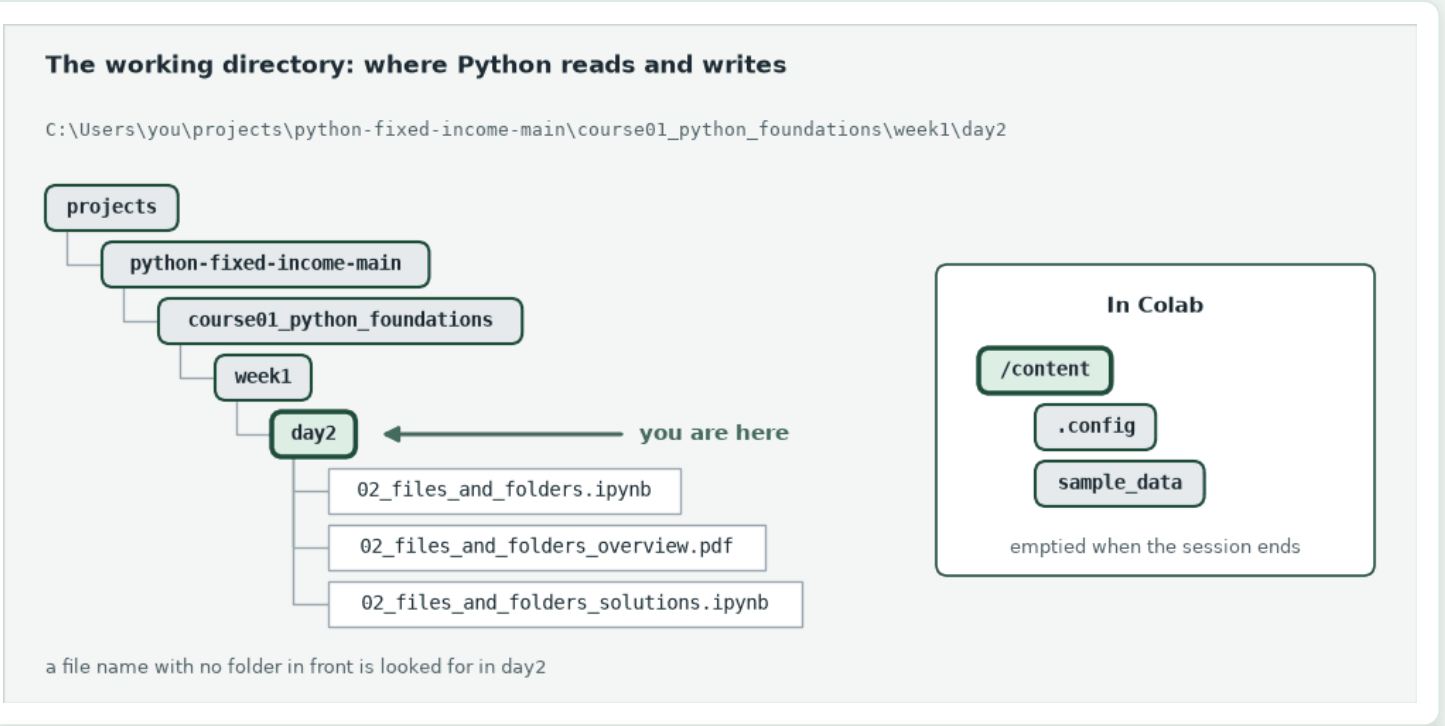
The book lives only while your notebook runs: restart it and the four bonds are gone. Give it a folder of its own, write it to a file there, and show me it reads back to 3,000,000 face and 2,958,750 dollars of cost.

The book in one cell: a new notebook starts empty

```
ticker_list = ['QVNE', 'DNMR', 'PLWK', 'HLVF']
coupon_list = [5.25, 6.125, 4.75, 7.5]
price_list = [98.5, 103.25, 95.0, 101.0]
face_list = [1500000, 500000, 750000, 250000]
print(f'The desk holds {len(ticker_list)} bonds and {sum(face_list):,} face')
The desk holds 4 bonds and 3,000,000 face
```

Today the book gets a folder, a file, and a way back in.

2.1 Where Python is working




```
import os
print(os.name)
nt
print(os.getcwd())
print(os.path.basename(os.getcwd()))
day2
```

nt means Windows; posix means macOS or Linux, Google Colab included.

Excel side by side

In desktop Excel, =CELL("filename",A1) shows a saved workbook's full path, file and sheet name at the end.

2.2 What the folder holds

```
names = os.listdir('.')
print(names)
print(len(names))
['02_files_and_folders.ipynb', '02_files_and_folders_overview.pdf', '02_files_and_folders_solutions.ipynb']
3
```

```
for name in names:
    print(name)
notebooks = [name for name in names if name.endswith('.ipynb')]
print(notebooks)
02_files_and_folders.ipynb
02_files_and_folders_overview.pdf
02_files_and_folders_solutions.ipynb
['02_files_and_folders.ipynb', '02_files_and_folders_solutions.ipynb']
```

listdir() returns a list of names, in no set order: here the notebook, its solutions, and its overview PDF.

2.3 A folder for the book

Start clean, then make the folder

```
import shutil
shutil.rmtree('desk_holdings', ignore_errors=True)
os.mkdir('desk_holdings')
print(os.path.exists('desk_holdings'))
True
```

Check first, so the cell is safe to rerun

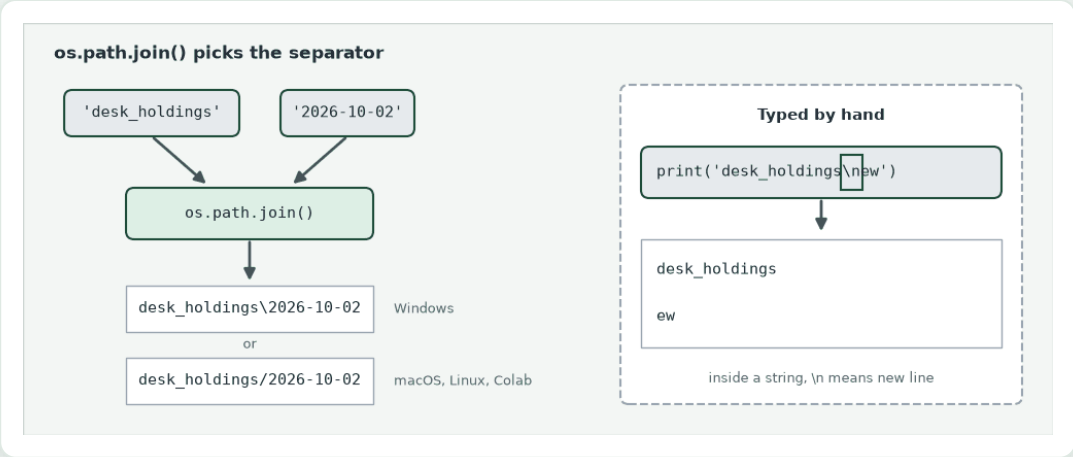
```
if not os.path.exists('desk_holdings'):
    os.mkdir('desk_holdings')
    print('Folder created')
else:
    print('Folder already exists')
Folder already exists
```

The same folder twice: Python refuses

```
os.mkdir('desk_holdings')
FileExistsError: [WinError 183] Cannot create a file when that file already exists: 'desk_holdings'
```

shutil.rmtree() deletes a folder and everything in it at once, with no recycle bin and no question: point it only at a folder your own code made.

Build a path, then type one by hand



os.path.join()

```
day_folder = os.path.join('desk_holdings', '2026-10-02')
print(day_folder)
os.mkdir(day_folder)
print(os.listdir('desk_holdings'))
desk_holdings\2026-10-02
['2026-10-02']
```

Typed by hand

```
print('desk_holdings\new')
desk_holdings
ew
```

A forward slash works on Windows too, so 'desk_holdings/new' is safe.

2.4 Write the book to a file

Open, write, close

```
file_path = os.path.join(day_folder, 'holdings.csv')
print(file_path)
file = open(file_path, 'w', encoding='utf-8')
file.write('Ticker,Coupon,Price,Face\n')
file.close()
desk_holdings\2026-10-02\holdings.csv
```



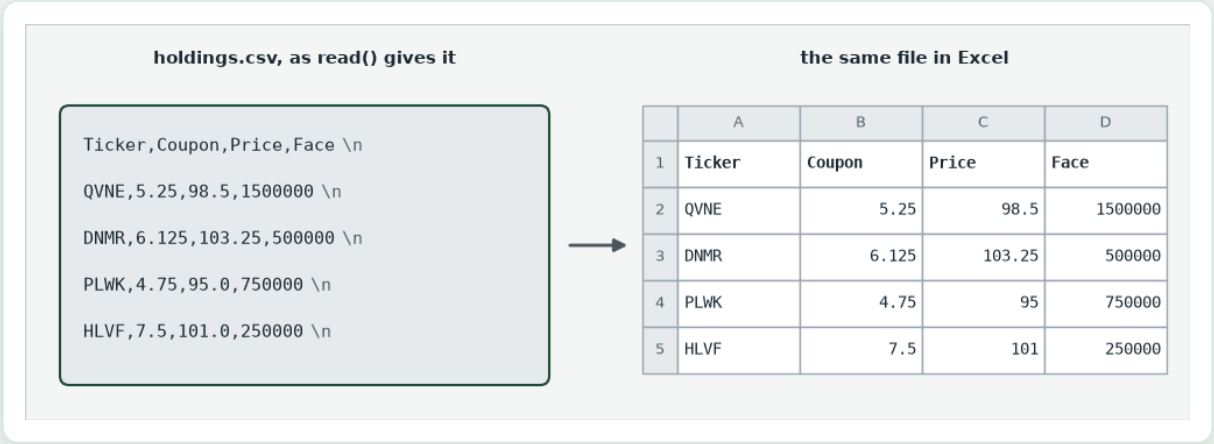
Let a loop write the bonds: 'a' adds to the end

```
with open(file_path, 'a', encoding='utf-8') as file:
    for ticker, coupon, price, face in zip(ticker_list, coupon_list, price_list, face_list):
        file.write(f'{ticker},{coupon},{price},{face}\n')
```

Always give encoding='utf-8': without it Windows uses an older default, and an accented letter or a currency sign can come back wrong on another machine.

Read it back

```
with open(file_path, 'r', encoding='utf-8') as file:
    print(file.read())
Ticker,Coupon,Price,Face
QVNE,5.25,98.5,1500000
DNMR,6.125,103.25,500000
PLWK,4.75,95.0,750000
HLVF,7.5,101.0,250000
```



Excel side by side

Double-click holdings.csv and Excel shows five rows and four columns, splitting each line at the commas. That holds where Windows' list separator is a comma (US and UK settings); where it is a semicolon, each line lands in one column.

2.5 Read the book back, line by line

A wrong name: holding.csv, one letter short

```
file = open(os.path.join(day_folder, 'holding.csv'), 'r', encoding='utf-8')
FileNotFoundError: [Errno 2] No such file or directory: 'desk_holdings\\2026-10-02\\holding.csv'
```

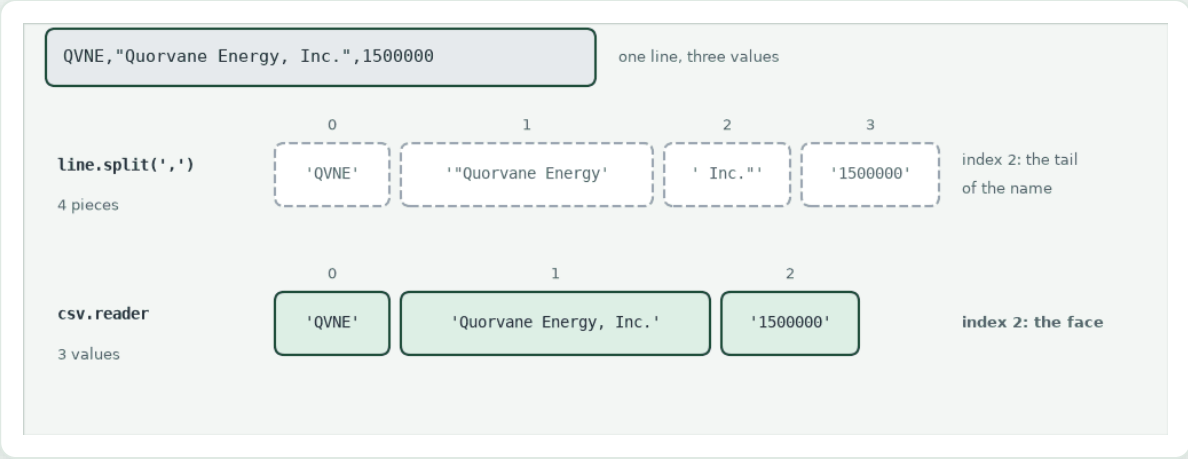
```
total_face = 0
for line in lines[1:]:
    values = line.strip().split(',')
    total_face += int(values[3])
print(f'The total face held is {total_face:,}')
The total face held is 3,000,000
```

Step	Gives
`line`	'QVNE,5.25,98.5,1500000\n', one string
`line.strip()`	'QVNE,5.25,98.5,1500000', the \n gone
`.split(',')`	['QVNE', '5.25', '98.5', '1500000'], four strings
`values[3]`	'1500000', still text: the TypeError
`int(values[3])`	1500000, a number to add

Excel side by side

split(',') is Text to Columns on Excel's Data tab, with the comma as the delimiter.

2.6 A comma inside a name




```
line = 'QVNE,"Quorvane Energy, Inc.",1500000'
values = line.split(',')
['QVNE', '"Quorvane Energy', ' Inc."', '1500000']
4 pieces; index 2 holds  Inc."
for values in csv.reader([line]):
['QVNE', 'Quorvane Energy, Inc.', '1500000']
3 values; index 2 holds 1500000
```

Read by column name: the book's cost from the file

```
total_cost = 0
with open(file_path, 'r', encoding='utf-8', newline='') as file:
    for row in csv.DictReader(file):
        total_cost += int(row['Face']) * float(row['Price']) / 100
print(f'The book cost ${total_cost:,.2f}')
The book cost $2,958,750.00
```

2.7 A path that knows itself

```
from pathlib import Path
holdings_path = Path('desk_holdings') / '2026-10-02' / 'holdings.csv'
print(holdings_path)
desk_holdings\2026-10-02\holdings.csv
csv_names = [path.name for path in holdings_path.parent.glob('*.csv')]
print(csv_names)
['holdings.csv']
```

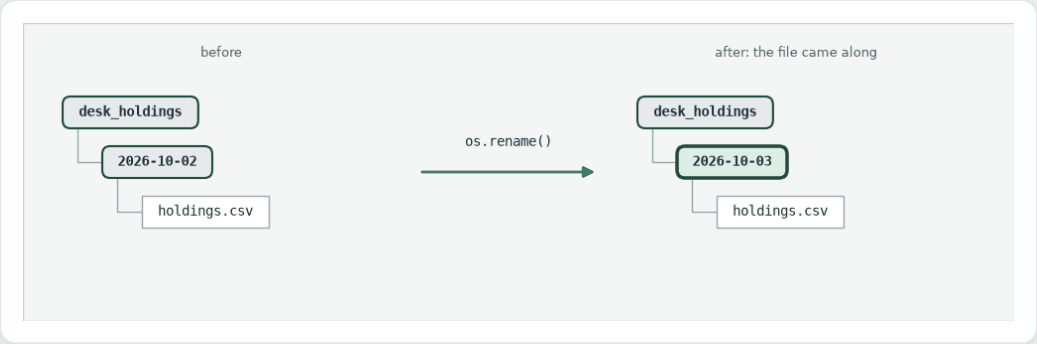
```
print(holdings_path.exists())
print(holdings_path.name)
print(holdings_path.parent)
print(holdings_path.parent.name)
True
holdings.csv
desk_holdings\2026-10-02
2026-10-02
```

Job	With os	With pathlib
Join the parts of a path	`os.path.join('desk_holdings', '2026-10-02')`	`Path('desk_holdings') / '2026-10-02'`
Check that it exists	`os.path.exists(path)`	`path.exists()`
Take the last part	`os.path.basename(path)`	`path.name`
Find the files that match	`os.listdir() with endswith()`	`path.glob('*.csv')`

Excel side by side

* is Excel's wildcard too: with file names in A1:A10, =COUNTIF(A1:A10,"*.csv") counts the names that end in .csv.

2.8 Move and tidy




```
new_day_folder = os.path.join('desk_holdings', '2026-10-03')
os.rename(day_folder, new_day_folder)
print(os.listdir('desk_holdings'))
print(os.listdir(new_day_folder))
['2026-10-03']
['holdings.csv']
```

Delete the file, then the folders: inner first, then outer

```
os.remove(os.path.join(new_day_folder, 'holdings.csv'))
print(os.listdir(new_day_folder))
[]
os.rmdir('desk_holdings')
OSError: [WinError 145] The directory is not empty: 'desk_holdings'
```

```
os.rmdir(new_day_folder)
os.rmdir('desk_holdings')
print(os.path.exists('desk_holdings'))
False
```

rmdir() refuses a folder that is not empty, a safety feature: the inner folder had to go first.

The note, answered

The note asked	Answer
A folder of its own	desk_holdings, a folder per day inside
Write it to a file	holdings.csv, four bonds by one loop
3,000,000 face read back	3,000,000, by split
2,958,750 dollars of cost read back	2,958,750.00, by csv.DictReader

Until today you typed the book; now it comes out of a file, the way real data reaches a desk.

On Day 3 errors become the lesson: you read a whole traceback and fix the line it points at.